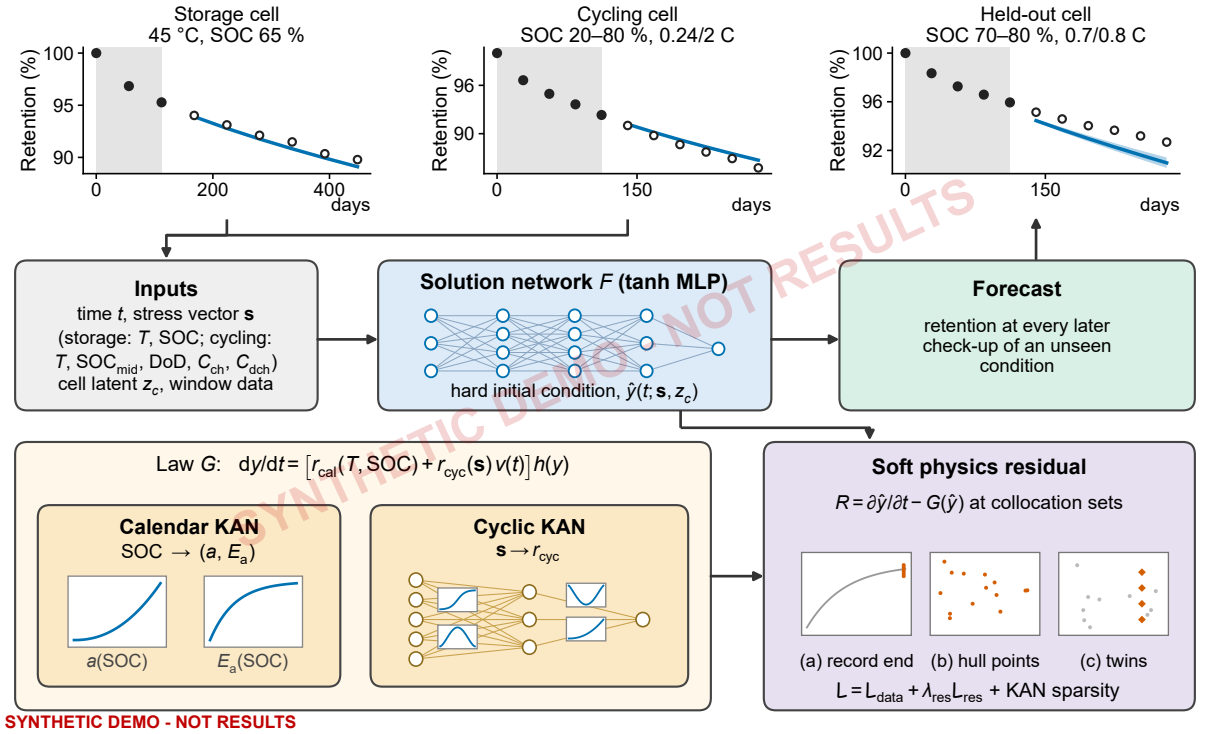

Storage cell
45 °C, SOC 65 %
Cycling cell
SOC 20–80 %, 0.24/2 C
Held-out cell
SOC 70–80 %, 0.7/0.8 C
Retention (%)
days
Inputs
time t, stress vector s
(storage: T, SOC; cycling:
T, SOC_mid, DoD, C_ch, C_dch)
cell latent z_c, window data
Solution network F (tanh MLP)
hard initial condition, ŷ(t; s, z_c)
Forecast
retention at every later
check-up of an unseen
condition
Law G: dy/dt = [r_cal(T, SOC) + r_cyc(s) v(t)] h(y)
Calendar KAN
SOC → (a, E_a)
a(SOC)
E_a(SOC)
Cyclic KAN
s → r_cyc
Soft physics residual
R = ∂ŷ/∂t − G(ŷ) at collocation sets
(a) record end
(b) hull points
(c) twins
L = L_data + λ_res L_res + KAN sparsity
SYNTHETIC DEMO - NOT RESULTS